

Change of basis

The same vector, described in someone else's language.

THE IDEA

Coordinates are not absolute — they depend on which basis vectors you use. Another person might use different basis vectors, so the *same arrow* gets different numbers. A **change of basis** translates between the two descriptions.

The translation rule. Let A be the matrix whose columns are *their* basis vectors written in *our* coordinates. Then:
their language $\rightarrow$ our language: multiply by A . Our language $\rightarrow$ their language: multiply by A^{-1} .

TRANSFORMING INSIDE THEIR LANGUAGE

To apply a transformation M while staying in their coordinate system, use $A^{-1} M A$ — read right to left: translate into our language, transform, translate back. Matrices related this way are called **similar**, and they describe the same underlying transformation seen from two viewpoints.

WORKED EXAMPLE

CHAPTER 13 — WORKED EXAMPLE

same vector, described with different basis vectors

THE TWO DIRECTIONS

their language → our language : multiply by A

our language → their language : multiply by A^{-1}

A = matrix whose columns are THEIR basis vectors (in our coords)

EXAMPLE — their coords → our coords

their basis : $b_1 = (2, 1)$, $b_2 = (-1, 1)$

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \quad \text{vector in their language : } (3, 2)$$

$$3 \times (2, 1) = (6, 3)$$

$$2 \times (-1, 1) = (-2, 2)$$

$$\text{add : } (6 + (-2), 3 + 2) = (4, 5)$$

same arrow — just described with our $\hat{i}$ and $\hat{j}$

Transformation in their language : $A^{-1} M A$

read right to left : into our language → transform → back to theirs

Why it matters for ML: choosing a good basis is often the whole trick. PCA finds a basis where the data's variance lines up with the axes; embeddings are learned bases where directions carry meaning.

TEST — Chapter 13

1. Their basis: $b_1 = (1, 2)$, $b_2 = (3, 0)$. Their vector $(2, 1)$. What is it in our coordinates? → **$(5, 4)$**
2. Their basis: $b_1 = (2, 0)$, $b_2 = (0, 4)$. Their vector $(3, 2)$. Ours? → **$(6, 8)$**
3. Their basis: $b_1 = (2, 1)$, $b_2 = (-1, 1)$. Their vector $(1, -2)$. Ours? → **$(4, -1)$**

FORMULAS FROM THIS CHAPTER

Their → ours

multiply by A

Ours → theirs	multiply by A^{-1}
A	columns = their basis vectors in our coords
Transform in their language	$A^{-1} M A$